

COURSE STRUCTURE

Course Code	BCA30140				
Course Category	Program Elective -I				
Course Title	Introduction to NoSQL Database				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	1	0		4
Credits	3	1	0		4
Assessment Code	TT1				

Prerequisites:

Knowledge of database concepts

Basic understanding of Big Data platforms

Course Objectives:

1. Understand importance of NoSQL Databases.

Course Outcomes:

On completion of the course, student will be able to–

- 1 Explain and compare different types of NoSQL Databases
- 2: Compare and contrast RDBMS with different NoSQL databases.
- 3: Demonstrate the detailed architecture and performance tune of Document-oriented NoSQL databases.
- 4: Explain performance tune of Key-Value Pair NoSQL databases.
- 5: Apply Nosql development tools on different types of NoSQL Databases.

Course Contents:

Unit 1: Introduction to NoSQL [9]

Overview and History of NoSQL Databases. Definition of the Four Types of NoSQL Database, The Value of Relational Databases, Getting at Persistent Data, Concurrency, Integration, Impedance Mismatch, Application and Integration Databases, Attack of the Clusters, The Emergence of NoSQL, Key Points.

Unit 2: Types of NoSQL Databases [9]

Comparison of relational databases to new NoSQL stores, MongoDB, Cassandra, HBASE, Neo4j use and

deployment, Application, RDBMS approach, Challenges NoSQL approach, Key-Value and Document Data Models, Column-Family Stores, Aggregate-Oriented Databases. Replication and sharding, MapReduce on databases. Distribution Models, Single Server, Sharding, Master-Slave Replication, Peer-to-Peer Replication, Combining Sharding and Replication.

Unit 3: Features and Application of NoSQL Databases [9]

NoSQL Key/Value databases using MongoDB, Document Databases, Document oriented Database Features, Consistency, Transactions, Availability, Query Features, Scaling, Suitable Use Cases, Event Logging, Content Management Systems, Blogging Platforms, Web Analytics or Real-Time Analytics, E-Commerce Applications, Complex Transactions Spanning Different Operations, Queries against Varying Aggregate Structure.

Unit 4: Column-oriented NoSQL DB [9]

Column- oriented NoSQL databases using Apache HBASE, Column-oriented NoSQL databases using Apache Cassandra, Architecture of HBASE, Column-Family Data Store Features, Consistency, Transactions, Availability, Query Features, Scaling, Suitable Use Cases, Event Logging, Content Management Systems, Blogging Platforms, Counters, Expiring Usage.

Unit 5: NoSQL Development Tools [9]

NoSQL Key/Value databases using Riak, Key-Value Databases, Key-Value Store, Key-Value Store Features, Consistency, Transactions, Query Features, Structure of Data, Scaling, Suitable Use Cases, Storing Session Information, User Profiles, Preferences, Shopping Cart Data, Relationships among Data, Multi operation Transactions, Query by Data, Operations by Sets. Graph NoSQL databases using Neo4, NoSQL database development tools and programming languages, Graph Databases, Graph Database. Features, Consistency, Transactions, Availability, Query Features, Scaling, Suitable Use Cases.

Learning Resources:

Reference Books:

Sadalage, P. & Fowler, NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence, Wiley Publications, 1st Edition, 2019.

Web Resources

Weblinks

1. <https://www.ibm.com/cloud/learn/nosql-databases>
2. <https://www.coursera.org/lecture/nosql-databases/introduction-to-nosql-VdRNp>
3. <https://www.geeksforgeeks.org/introduction-to-nosql/>
4. <https://www.javatpoint.com/nosql-databa>

MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc

Pedagogy:

- Participative Learning,
- Discussion
- Demonstrations
- Practical
- Assignment